

>saturnCFD

saturnCFD v1

MPI and GPU finite-volume CFD with validated methodology

Ananth Narayan

Indian Institute of Technology Madras

31 October 2026

Outline

1. Motivation and scope
2. What is implemented
3. Mesh and configuration
4. Parallel model
5. Finite-volume method
6. Verification and validation
7. Commands and tests
8. Status and roadmap

What saturnCFD is

Goal

A single finite-volume code for scalar transport and incompressible flow, portable across CPU and GPU, parallel through MPI, driven entirely by YAML, and verified before it is trusted.

- Structured hexahedral mesh imported from Gmsh `.msh`.
- 3D from the start; a one-cell-thick mesh is treated as 2D.
- Solvers: Burgers, heat/convection–diffusion, Stokes, incompressible Navier–Stokes.
- Kokkos for CPU/GPU; MPI for distributed memory.
- Every numerical choice is documented, cited and tested.

v0 is preserved on `v0-archive`. v1 is a deliberate rewrite, not a patch on the old AMR container.

Why this order?

1. Burgers: nonlinear convection, shocks, limiters, diffusion.
2. Heat: scalar transport, Péclet effects, boundary layers.
3. Stokes: vector elliptic operator, pressure coupling, linear solver.
4. Incompressible Navier–Stokes: all of the above together.

Idea

Each PDE tests one more ingredient. A failure is localised before the next solver is added.

Compressible flow, AMR and two-phase flow are later milestones.

Implemented substrate

- Gmsh 2.2 reader; structured-hex classification and `checkMesh` report.
- Default `.vtu` + `.json`; optional `-writeVTK` patches and `.pvd`.
- yaml-cpp configuration with a typed schema and generated templates.
- SI-only validation: non-SI input is rejected, not converted.
- Single-dash CLI with help topics and shell completion.
- MPI Cartesian partition and 26-neighbour halo exchange.
- Kokkos `CellField`/`FaceField`, device pack/unpack halo, structured `LocalMesh`.
- Interpolation, Gauss gradient, divergence, non-orthogonal Laplacian and Dirichlet/Neumann ghosts.
- Upwind/Rusanov/MUSCL convection, TVD limiters and advection–diffusion MMS.
- Divergence theorem on a sheared mesh; simplified Rhie–Chow flux.
- PETSc 3.19.6 through `saturnLinAlg`; 1D/3D Poisson and all-Neumann nullspace tests.
- Solver interface, registry, `TimeControl` and explicit Simulation loop (forward Euler, SSPRK2/3).
- 1D/2D/3D `saturnBurger`: MUSCL/Rusanov flux, diffusion, Dirichlet and periodic boundaries.
- 1D/2D/3D `saturnHeat`: prescribed velocity, convection–diffusion, periodic Fourier validation.
- `output.vtkEvery`: `.vtu` frames plus a ParaView `.pvd` collection.
- Periodic Laplacian, gradient and divergence MMS at order 2 on 1/2/4 ranks.
- 46 CTest cases, regression tests, Doxygen HTML/PDF.

Architecture

Layer	Responsibility
`core`	types, SI units, errors, timer
`config`	yaml-cpp, schema, validation, templates
`mesh`	Gmsh, `checkMesh`, structured view, VTK
`comm`	partition, ghost ownership, halo exchange
`fv`	fields, gradient, divergence, Laplacian, fluxes
`linalg`	MPI PETSc systems
`simulation`	solver registry, time loop, diagnostics
`solvers`	Burgers, heat, Stokes, incompressible

A solver owns its equations. It does not own the mesh, the halo, the YAML parser or the linear algebra.

Structured mesh and `checkMesh`

- Gmsh 2.2 ASCII: points, hexahedra, boundary physical groups.
- Classifier walks the cell dual graph and assigns a contiguous (i, j, k).
- Rejects tetrahedra, non-manifold faces, negative volumes and non-contiguous hex meshes.
- Reports counts, patches, volume/area ranges, non-orthogonality, skewness, aspect ratio, structured flag and dimension.

2D rule

A one-cell-thick 3D mesh with `empty` patches is 2D. One internal representation, one solver path.

Gmsh [5]; OpenFOAM's `checkMesh` convention is a methodology reference.

YAML and SI units

```
solver: common
units:
  system: SI
mesh:
  file: channel.msh
time:
  end: 1.0
  cfl: 0.4
```

- yaml-cpp parses; the schema validates.
- Unknown keys are errors.
- `-dumpConfig` generates the template from the same registration the solver reads.
- `-checkConfig` runs all validation before allocation.

SI only in v1

`units.system: SI` or no block is accepted. `CGS`, `length: mm`, etc. are hard errors. A future units layer converts at the configuration boundary, never in a kernel.

MPI decomposition

global grid	$n_x \times n_y \times n_z$
process grid	$p_x \times p_y \times p_z$
local block	$[i_0, i_1] \times [j_0, j_1] \times [k_0, k_1]$

- `MPI_Dims_create` chooses the process grid; `MPI_Cart_create` handles periodicity.
- 2D uses $p_z = 1$; 1D uses $p_y = p_z = 1$.
- Remainders are distributed one cell at a time from the low index.
- No METIS: a structured mesh does not need a graph partition.

Why Cartesian blocks?

Deterministic neighbours, compact messages and a direct cell $\leftrightarrow$ rank mapping.

No shared cells: owner and ghost

- Every global cell has **one owner rank**.
- Only the owner writes the valid value.
- The neighbour keeps a **read-only ghost copy**.
- Two ghost layers support MUSCL/TVD.
- 26-neighbour exchange: faces, edges, corners.

Shared face, not shared cell

Both ranks compute the same face flux from the same valid/ghost pair. Each rank updates only its own cell. No double counting.

Halo order: local periodic copies → packed MPI → physical boundary conditions.

CPU and GPU execution

- Kokkos selects the backend at **build** time: Serial/OpenMP or CUDA/HIP.
- CPU and GPU are separate build directories: `buildCPU` and `buildGPU`.
- `mpirun` launches MPI ranks; the compiled backend decides whether a rank uses a CPU or a GPU.
- Performance model: **one rank per GPU**.
- One GPU: `mpirun -np 1`. Four GPUs: `mpirun -np 4`.
- Halo first implementation: device pack $\rightarrow$ host MPI $\rightarrow$ device unpack.

Case files are hardware-independent

CPU/GPU is not in `case.yaml`. Physics stays portable.

Collocated cell-centred FVM

- Velocity and pressure at cell centroids.
- Face fluxes reconstructed from the two adjacent cells.
- Rhie–Chow correction suppresses checkerboard pressure.
- Coupled saddle-point [1, 4] for steady Stokes.
- SIMPLE/SIMPLEC remain optional segregated paths for Navier–Stokes; PISO for transient flow.

Why not MAC?

Staggered MAC gives exact discrete continuity, but it splits the velocity components onto different meshes and fights the vector form, `.msh` import and the 2D/3D rule. Collocated FV matches OpenFOAM and the default Basilisk/Nektar++ approach [14, 2]; Rhie–Chow [10] restores pressure–velocity stability.

Governing equations

$$\text{Burgers:} \quad \frac{\partial u}{\partial t} + \nabla \cdot (u\mathbf{a}) = \nu \nabla^2 u + s$$

$$\text{Heat:} \quad \frac{\partial T}{\partial t} + \nabla \cdot (\mathbf{u} T) = \nabla \cdot (\alpha \nabla T) + S$$

$$\text{Stokes:} \quad -\nabla p + \mu \nabla^2 \mathbf{u} = \mathbf{f}, \qquad \nabla \cdot \mathbf{u} = 0$$

$$\text{Incompressible NS:} \quad \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \frac{\mathbf{f}}{\rho},$$
$$\nabla \cdot \mathbf{u} = 0$$

Projection/pressure correction from Chorin [3]; SIMPLE/PISO from Patankar [9] and Issa [7]; Rhie–Chow [10]; coupled saddle point from Benzi et al. [1] and Elman et al. [4].

Vector form and linear algebra

Discrete momentum

$$\mathbf{M}\frac{d\mathbf{u}}{dt} + \mathbf{C}(\mathbf{u})\mathbf{u} = -\mathbf{G}p + \mathbf{D}\mathbf{u} + \mathbf{f}$$

- One vector unknown per cell: u, v, w.
- One sparsity pattern; block size 3 or three right-hand sides.
- PETSc: `Mat`, `Vec`, `KSP`, `PC`.
- Pressure Poisson: CG + GAMG/HYPRE; Neumann null space handled explicitly.
- Coupled Stokes/NS: `MPIAIJ` with (u, v, w, p) per cell, GMRES/FGMRES + `PCFIELDSPLIT` Schur preconditioner.
- Velocity block GAMG/ILU; pressure block GAMG/HYPRE; nullspace attached with `MatNullSpace`.
- No home-grown Krylov solver.

Stokes: coupled saddle point

$$\begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ p \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

- One PETSc solve for velocity and pressure; no SIMPLE outer iteration for Stokes.
- Collocated FV + Rhie–Chow [10] stabilisation.
- Schur factorisation $S = -BA^{-1}B^T$; approximate A^{-1} and S^{-1} with AMG/ILU; GMRES handles the coupling [1, 4].
- Pressure nullspace attached for all-Neumann cases; mean pressure reported separately.
- The same block is reused inside Navier–Stokes Picard/Newton or time-step iterations.

Why coupled?

Stokes is linear and steady; pressure is a Lagrange multiplier. A coupled solve avoids under-relaxation and pseudo-time tuning.

Stokes validation plan

Case	Quantity	Reference
Poiseuille 2D/3D	velocity profile	analytic parabola
Couette 2D/3D	velocity profile	analytic line
MMS 2D/3D	velocity/pressure order	manufactured
Exit boundary	3D outflow	Sankhla & Das [13]
Nullspace	mean pressure	constant mode

Status

Poiseuille and Couette pass in 2D/3D; 2D runs on 1/2/4/8 ranks and 3D on 1/2/4. Fieldsplit (FGMRES + GAMG) is the default and matches direct to the linear-solve tolerance. The periodic MMS is source-consistent and second order in velocity (2.01 in 2D, 2.03 in 3D); the exit-boundary case follows.

Stokes: first validated results

Case	mesh	L^2	note
Poiseuille 2D	16^2	4.88×10^{-4}	second order
Poiseuille 2D	32^2	1.22×10^{-4}	second order
Poiseuille 2D	64^2	3.05×10^{-5}	example
Couette 2D	64^2	7.5×10^{-13}	exact line
Poiseuille 3D	$24^2 \times 8$	2.17×10^{-4}	extruded
Couette 3D	$24^2 \times 8$	5.7×10^{-12}	exact line

Output and validation

One VTU per case contains all four fields `u,v,w,p`; the example script validates both Poiseuille and Couette and draws 2D fields or 3D velocity-profile surfaces (numerical, analytic, error). The default fieldsplit uses GMRES + GAMG on velocity and preonly + GAMG on pressure, with a `selfp` Schur complement; direct LU remains the reference.

Manufactured solutions: why

Question MMS answers

Does the assembled discrete operator converge to the PDE at the expected order?

- Unit tests check one function; regression checks one accepted result.
- MMS checks the full operator, including faces, ghosts, BCs, geometry and partition boundaries.
- It catches sign/coefficient errors, missed faces and halo bugs that are invisible on one rank.

Salari & Knupp [12]; Roache [11]; Oberkampf & Roy [8].

Manufactured solutions: method

Choose a smooth ϕ, substitute into the PDE, and use the residual as the source:

$$s = \mathcal{L}(\phi).$$

For the periodic Laplacian:

$$\phi = \sin(2\pi x)\sin(2\pi y)\sin(2\pi z), \qquad s = -3(2\pi)^2\phi.$$

Refine h and compute

$$p \approx \frac{\log(e_{\text{coarse}}/e_{\text{fine}})}{\log 2}.$$

Expected orders

Central Laplacian/interpolation: 2. Upwind: 1. TVD on a smooth non-extremum: about 2, with local reduction at extrema.

Measured on 1, 2 and 4 ranks

Laplacian: 3D 1.978/1.994, 2D 1.994/1.999. Gradient/divergence: 1.967/1.992. Upwind: 0.960/0.990. MUSCL: 1.61/1.57 (minmod), 1.58/1.52 (van Leer), 1.15/1.42 (superbee). The 32^3 reference $L^2 = 0.1343558076$ is stored.

Validation roadmap

Solver	First validation	Reference
Burgers	viscous traveling wave	exact Cole–Hopf solution
Burgers	inviscid shock	characteristics/conservation
Heat	1D boundary layer	exact solution
Heat	2D/3D MMS	manufactured order
Stokes	Poiseuille/Couette 2D/3D	exact parabolic/linear
Stokes	MMS 2D/3D	manufactured order
Stokes	exit boundary	Sankhla & Das (2025)
Incompressible	cavity $Re = 100, 1000$	Ghia et al. [6]
Incompressible	Taylor–Green	energy decay/order

Rule

MMS and analytic tests before a physical benchmark. Observations are recorded, including failures and order reduction.

1D scalar Burgers: `saturnBurger`

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) = \nu \frac{\partial^2 u}{\partial x^2} + s$$

- Rusanov/local Lax–Friedrichs flux with canonical face orientation.
- MUSCL/TVD reconstruction; minmod, van Leer, superbee.
- Shared non-orthogonal Laplacian for diffusion.
- Explicit SSPRK3; Dirichlet and periodic boundaries; time-dependent exact boundary data.
- Registered in the CLI: `saturnCFD case.yaml`, `-listSolvers`, `-dumpConfig saturnBurger`.
- Tests: traveling wave, constant field, periodic inviscid shock.

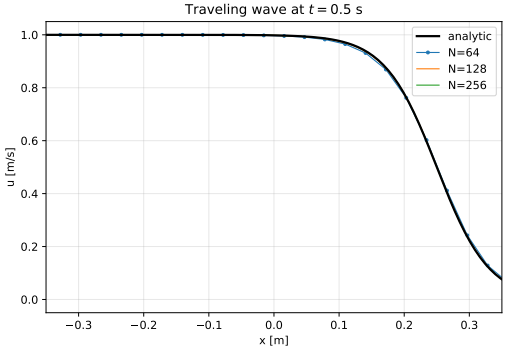
Measured

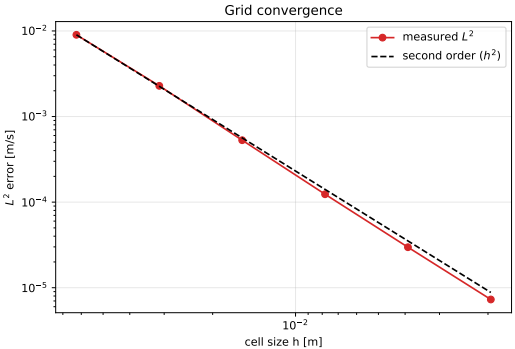
Traveling wave, $N = 32 \to 1024$, $\nu = 0.02$: L2 $9.0 \times 10^{-3} \to 7.3 \times 10^{-6}$, orders 1.98/2.11/2.10/2.06/2.03. At $N = 256$, 1/2/4 ranks give identical norms. 2D diagonal wave: L2 $9.9 \times 10^{-3} \to 6.0 \times 10^{-4}$, orders 1.96/2.10. Shock bounds $[-0.9954, 0.9954]$, integral 1.5×10^{-16}.

Why first

Exact solution, nonlinear convection, limiters, time integration, boundary ghosts, conservation and MPI

Burgers validation against the exact traveling wave

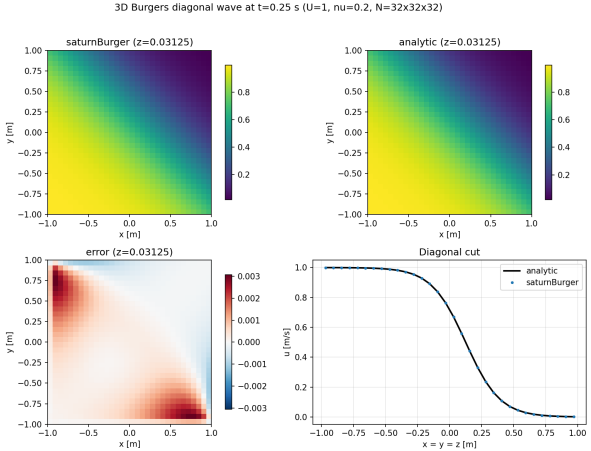




Verdict

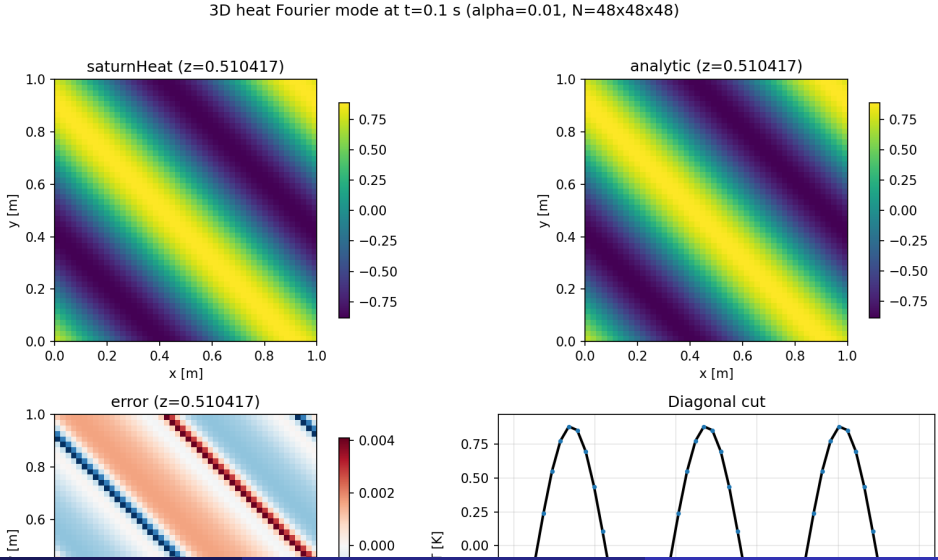
Second-order convergence to the analytic solution, a bounded and conservative shock, and rank-independent norms. The residual error at a finite grid is the expected truncation error, not a solver defect.

3D Burgers validation and restart



- 3D conservative flux $F(u)=(u^2/2,u^2/2,u^2/2)$.
- Exact diagonal plane wave.
- 32^3 example: $L^2 = 8.97 \times 10^{-4}$.
- Coarse 3D plane-wave order 1.37; dense cluster run next.
- `output.checkpointEvery`: binary restart files.
- `restart.file`: resumes at the checkpoint step/time.

3D heat validation: periodic Fourier mode



Build and run

```
# CPU + GPU into one install root
tools/compileSaturnCFD both
#  ~/.local/saturnCFD/{cpu,gpu}
#  ~/.local/bin/saturnCPU, saturnGPU

# run from any directory
saturnCPU /path/to/case.yaml
saturnGPU /path/to/case.yaml
saturnCFD /path/to/case.yaml   # CPU default
```

```
# threads
saturnCPU -threads 1 case.yaml
mpirun -np 4 saturnCPU \
  -threads 6 case.yaml

# GPUs: one rank per GPU
mpirun -np 1 saturnGPU case.yaml
mpirun -np 2 saturnGPU case.yaml

# tests
ctest --preset solver
ctest --preset saturnCFD
ctest --preset gpu
```

The installed binaries carry their backend. `KOKKOS_ROOT` is only needed to configure/build the GPU preset. Test binaries: `testSaturn`, `testSaturnMPI`, `testSaturnFV`, `testSaturnPETSc`.

Field output: VTK every N steps

```
output:
  prefix: results/burgers/run
  csv: true
  json: true
  vtkEvery: 10   # 0 = off
```

```
results/burgers/
  run_000000.vtu
  run_000010.vtu
  ...
  run_<final>.vtu
  run.pvd        # ParaView
```

What is written

The scalar cell field `u` at step 0, every `N` accepted steps and the final state. Open the `.pvd` collection in ParaView to animate the sequence. The VTK writer is shared with the mesh `-writeVTK` path.

Host requirements

```
# pre-flight check
python3 tools/checkRequirements.py

# install missing apt packages
python3 tools/checkRequirements.py \
  --install

# same check through CMake
cmake --build buildCPU --target doctor
```

- `requirements.toml` lists core, docs, editor and GPU tools with minimum versions.
- `checkRequirements.py --install`: PETSc, CUDA toolkit on NVIDIA hosts, and missing Kokkos builds.
- yaml-cpp 0.8 and doctest 2.4.11 are fetched by CMake.
- Missing optional tools are reported, not fatal.

Uniform names

`saturnCore`, `saturnLinAlg`, `saturnSimulation`, `saturnBurger`; `testSaturn`, `testSaturnMPI`, `testSaturnFV`, `testSaturnPETSc`. Doxygen HTML shows its build date (`TIMESTAMP = YES`).

Help, clean and completion

```
saturnCFD -help
saturnCFD -help build
saturnCFD -help gpu
saturnCFD -help mpi
saturnCFD -help petsc
saturnCFD -help clangd
saturnCFD -help test
saturnCFD -help clean

source <(saturnCFD -completion bash)

cmake --build buildCPU --target compileDb
clangd --compile-commands-dir=buildCPU

cmake --build buildCPU --target clean
rm -rf buildCPU buildGPU
```

No memorisation

Test presets: `default`, `cpu`, `gpu`, `solver`, `saturnCFD`. CLI: `saturnCFD -help`. Build: `-help build`. clangd: `-help clangd` and `compileDb`. Clean: `-help clean`.

Current test status

Suite	Result
CTest total	46/46 passing
MPI distribution & halo	1, 2, 4 ranks passing
Kokkos field + halo	2 ghosts, 26 directions, 1/2/4 ranks
LocalMesh + operators	linear exactness, 1/2/4 ranks
Non-orthogonal Laplacian	sheared linear field exact
MMS Laplacian	order 2 in 2D/3D, 1/2/4 ranks
MMS gradient/divergence	order 1.97/1.99, 1/2/4 ranks
MMS upwind/advection–diffusion	order 0.96 and 0.85–0.94
MMS MUSCL	order 1.15–1.61, TVD step bounded
Divergence theorem	sheared mesh, roundoff
Rhie–Chow	linear zero, checkerboard suppressed
PETSc Poisson/nullspace	order 2, 1/2/4 ranks
Solver loop	constant, diffusion decay, bounded advection
`saturnBurger`	1D N=32–1024 order 2.03; 2D order 2.10
`saturnHeat`	3D Fourier order 2.28/2.44; 2D 2.37/2.20
VTK export	1D frames + PVD; CTest `vtk`
Mesh `checkMesh`	pass/reject fixtures passing
Config SI-only	accept/reject fixtures passing

Status and next steps

Done

- Mesh + `checkMesh`
- YAML schema + SI validation
- CLI/help/completion/install
- MPI partition + 26-neighbour halo
- rank-0 mesh load + local geometry broadcast
- Kokkos field + device halo
- `LocalMesh` + operators + BC ghosts
- Non-orthogonal Laplacian + upwind/MUSCL convection
- TVD limiters, divergence theorem, Rhie–Chow flux
- Laplacian/gradient/divergence/advection MMS

Next

- FV coupled matrix assembly
- `saturnStokes`: 2D/3D Poiseuille/Couette validation
- Implicit BE/CN/BDF2
- Incompressible Navier–Stokes
- CUDA build verification
- Cluster scaling study

Take-away

What makes this code trustworthy

1. A structured, validated mesh and a strict YAML/SI boundary.
2. An ownership model with no shared cells and a tested halo.
3. A single source for CPU/GPU through Kokkos, not two codes.
4. Verification by manufactured solutions before benchmarks.
5. A CLI and test layout that a user can discover, not memorise.

Questions and discussion

References I


M. Benzi, G. H. Golub, and J. Liesen.
Numerical solution of saddle point problems.
Acta Numerica, 14:1–137, 2005.


C. D. Cantwell, D. Moxey, A. Comerford, A. Bolis, G. Rocco, G. Mengaldo, D. De Grazia, S. Yakovlev, J.-E. Lombard, D. Ekelschot, B. Jordi, H. Xu, Y. Mohamied, C. Eskilsson, B. Nelson, P. Vos, C. Biotto, R. M. Kirby, and S. J. Sherwin.
Nektar++: An open-source spectral/hp element framework.
Computer Physics Communications, 192:205–219, 2015.


A. J. Chorin.
Numerical solution of the navier–stokes equations.
Mathematics of Computation, 22(104):745–762, 1968.


H. C. Elman, D. J. Silvester, and A. J. Wathen.
Finite Elements and Fast Iterative Solvers: with Applications in Incompressible Fluid Dynamics.
Oxford University Press, 2 edition, 2014.

References II

C. Geuzaine and J.-F. Remacle.
Gmsh: a three-dimensional finite element mesh generator with built-in pre- and post-processing facilities.
International Journal for Numerical Methods in Engineering, 79(11):1309–1331, 2009.

U. Ghia, K. N. Ghia, and C. T. Shin.
High-Re solutions for incompressible flow using the navier–stokes equations and a multigrid method.
Journal of Computational Physics, 48(3):387–411, 1982.

R. I. Issa.
Solution of the implicitly discretised fluid flow equations by operator-splitting.
Journal of Computational Physics, 62(1):40–65, 1986.

W. L. Oberkampf and C. J. Roy.
Verification and Validation in Scientific Computing.
Cambridge University Press, 2010.

References III

S. V. Patankar.
Numerical Heat Transfer and Fluid Flow.
Hemisphere Publishing, 1980.

C. M. Rhie and W. L. Chow.
Numerical study of the turbulent flow past an airfoil with trailing edge separation.
AIAA Journal, 21(11):1525–1532, 1983.

P. J. Roache.
Code verification by the method of manufactured solutions.
Journal of Fluids Engineering, 124(1):4–10, 2002.

K. Salari and P. Knupp.
Code verification by the method of manufactured solutions.
Technical Report SAND2000-1444, Sandia National Laboratories, 2000.

References IV


K. Sankhla and S. K. Das.
Solution of three-dimensional steady state Stokes flow problems with exit boundary conditions by the boundary element method.
Computers & Mathematics with Applications, 195:1–13, 2025.


H. G. Weller, G. Tabor, H. Jasak, and C. Fureby.
A tensorial approach to computational continuum mechanics using object-oriented techniques.
Computers in Physics, 12(6):620–631, 1998.